

HW 5

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IISC - CFD

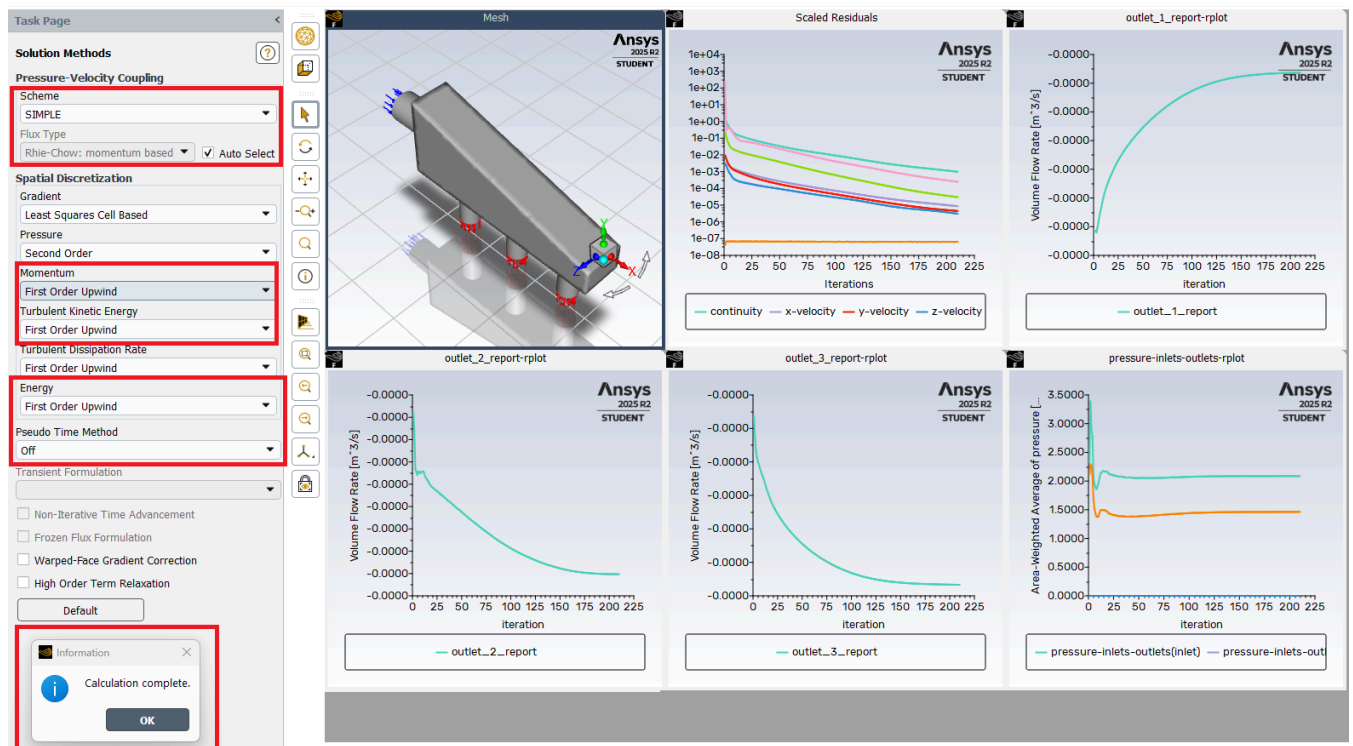
Input parameters used for the model and calculation are provided at the end of the document. Momentum is calculated using the formula below used as a user defined function.

$$M = \rho * \sqrt{(V_x^2 + V_y^2 + V_z^2)}$$

Observation : The scaled residuals decrease with iterations and reach a stable plateau. The outlet probe values (momentum and pressure) also stabilize, indicating that the solution has converged.

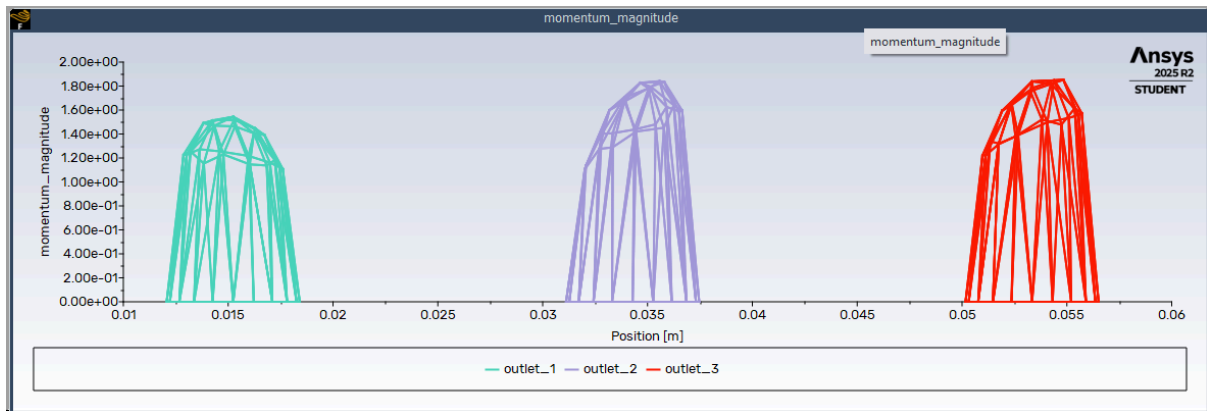
First order upwind

Convergence

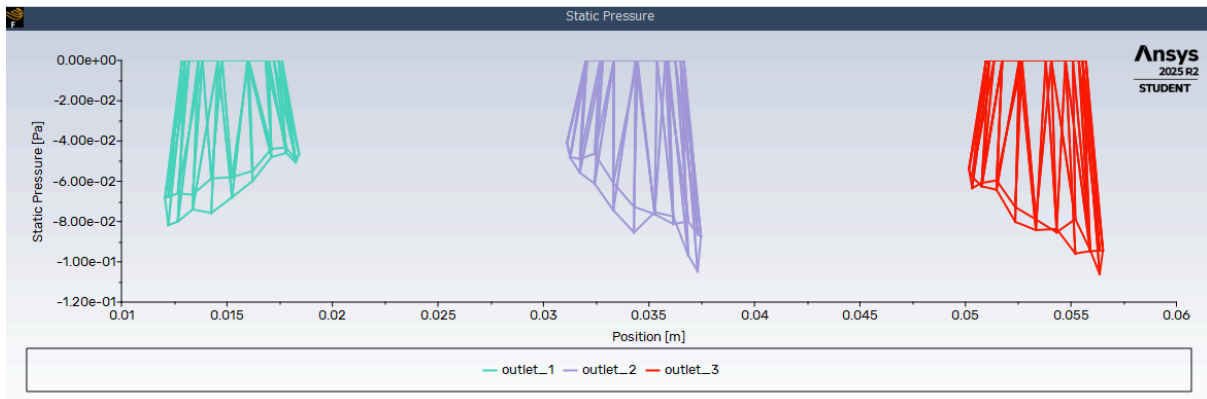


Probe plots

Momentum

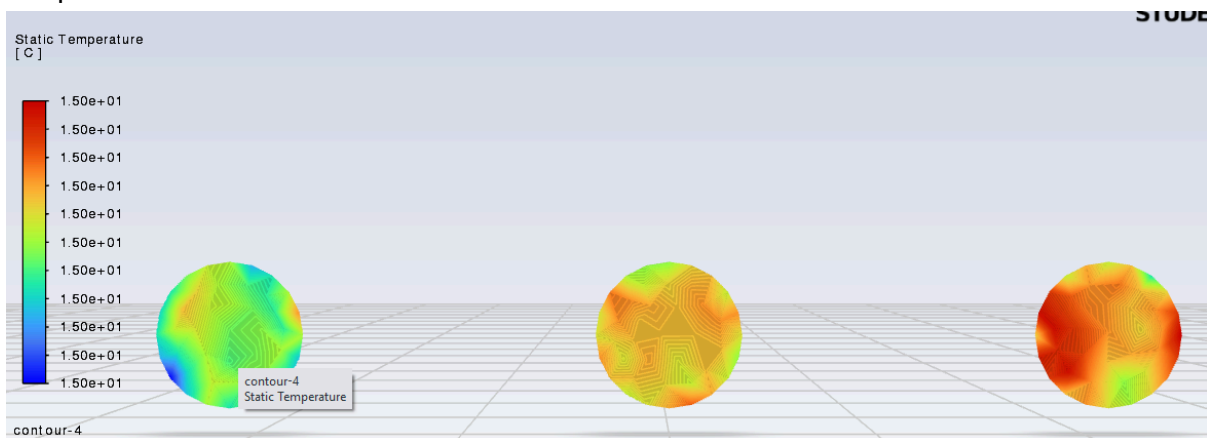


Pressure

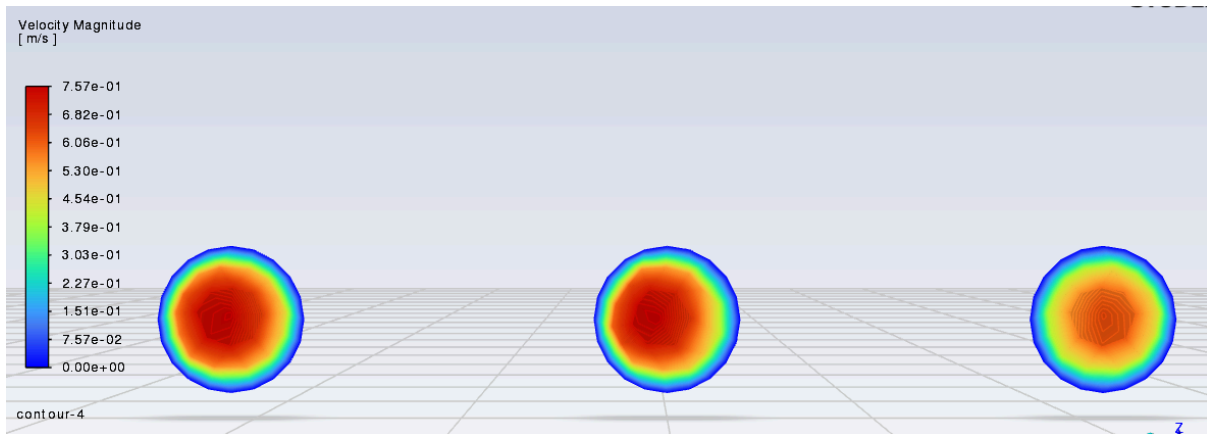


Contours

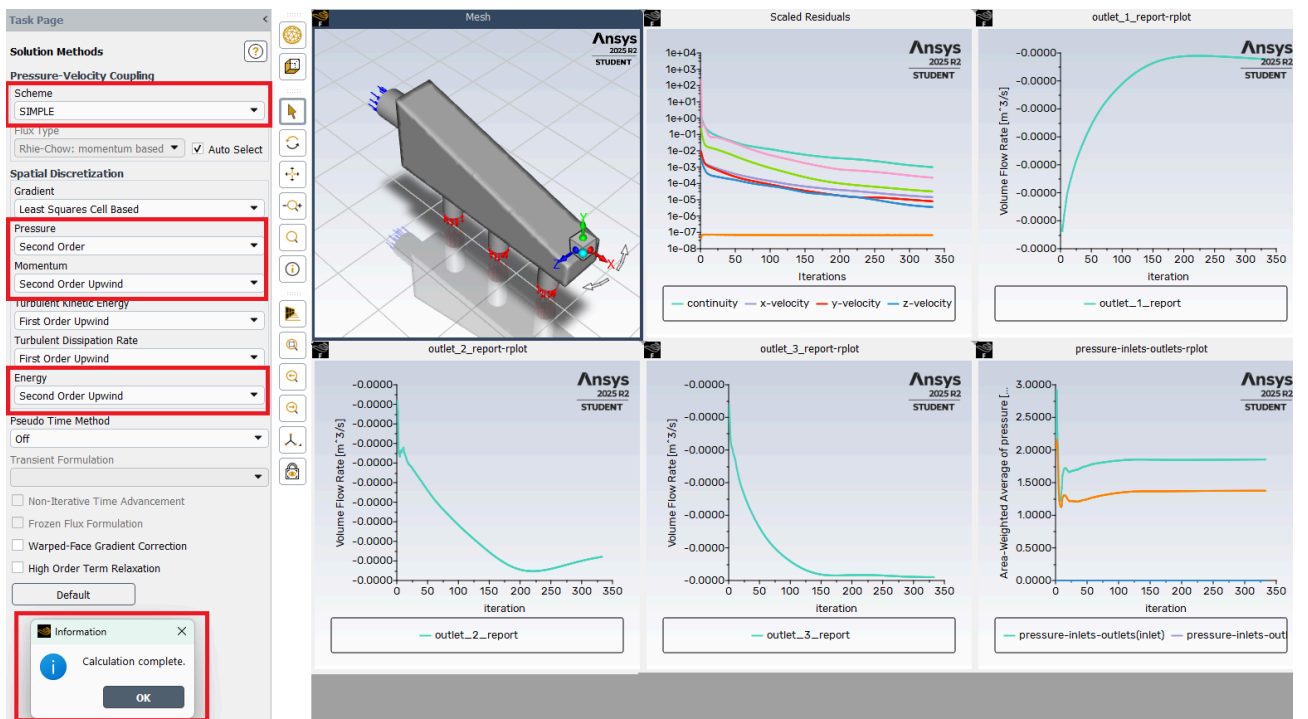
Temperature



Velocity



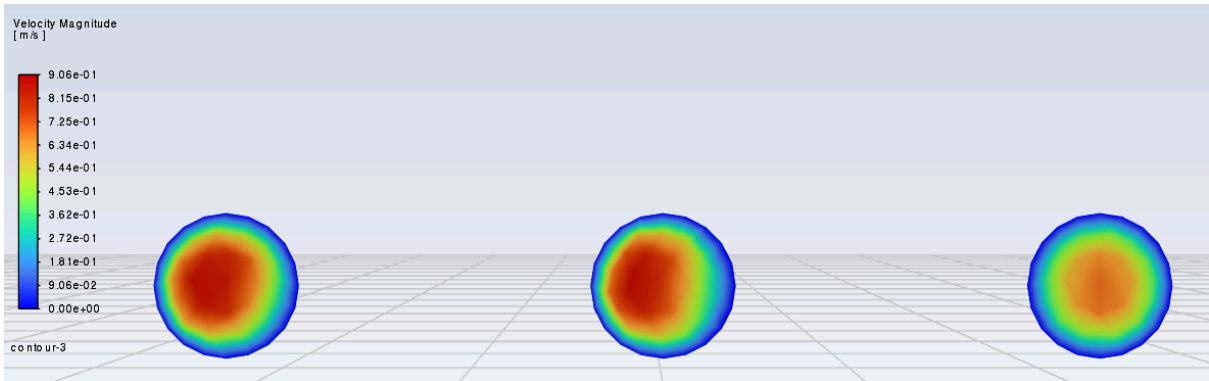
Second order



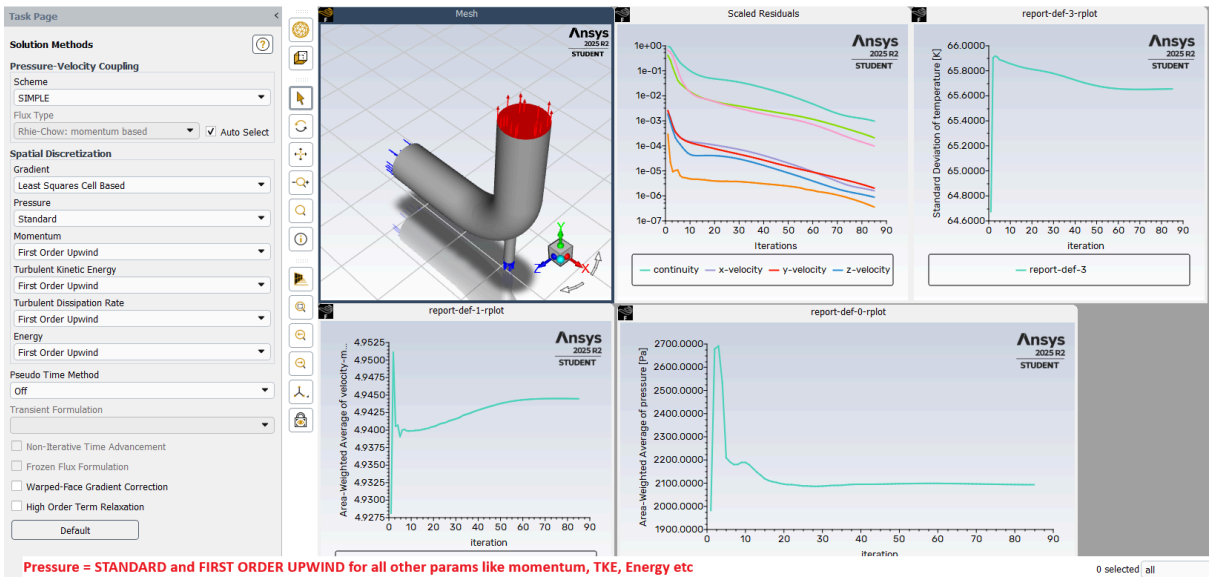
Probe plots

Momentum

Temperature



Example Mixing junction



The input summary used for the test is as below.

Fluent

Version: 3d, pbns, rke (3d, pressure-based, realizable k-epsilon)

Release: 25.2.0

Title:

Models

Model	Settings
Space	3D
Time	Steady
Viscous	Realizable k-epsilon turbulence model
Wall Treatment	Enhanced Wall Treatment
Heat Transfer	Enabled
Solidification and Melting	Disabled
Radiation	None
Species	Disabled

Coupled Dispersed Phase	Disabled
NOx Pollutants	Disabled
SOx Pollutants	Disabled
Soot	Disabled
Mercury Pollutants	Disabled
Structure	Disabled
Acoustics	Disabled
Eulerian Wall Film	Disabled
Potential/Electrochemistry	Disabled
Multiphase	Disabled

Material Properties

Material: air (fluid)

Property	Units	Method	Value(s)
Density	kg/m ³	constant	1.225
Cp (Specific Heat)	J/(kg K)	constant	1006.43
Thermal Conductivity	W/(m K)	constant	0.0242
Viscosity	kg/(m s)	constant	1.7894e-05

Material: aluminum (solid)

Property	Units	Method	Value(s)
Density	kg/m ³	constant	2719
Cp (Specific Heat)	J/(kg K)	constant	871
Thermal Conductivity	W/(m K)	constant	202.4

Cell Zone Conditions

Zones

name	id	type	material
fff_solid	3	fluid	air

Setup Conditions

fff_solid	
Condition	Value
Material Name	air
Frame Motion?	no
Mesh Motion?	no

Boundary Conditions

Zones

name	id	type

wall-fff_solid	1	wall
inlet	6	velocity-inlet
outlet_1	7	pressure-outlet
outlet_2	8	pressure-outlet
outlet_3	9	pressure-outlet

Setup Conditions

wall-fff_solid

Condition	Value

Material Name	aluminum
Thermal BC Type	Heat Flux
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip

inlet

Condition	Value

Velocity Magnitude [m/s]	1.5
Temperature [C]	15

outlet_1

Condition	Value

outlet_2

Condition	Value

outlet_3

Condition	Value

Solver Settings

Equations

Equation	Solved

Flow	yes
Turbulence	yes
Energy	yes

Numerics

Numeric	Enabled

Absolute Velocity Formulation	yes

Under-Relaxation Factors

Variable	Relaxation Factor

Pressure	0.7
Density	1

Body Forces	1
Momentum	0.7
Turbulent Kinetic Energy	0.8
Turbulent Dissipation Rate	0.8
Turbulent Viscosity	1
Energy	1

Linear Solver

Variable	Solver Type	Termination Criterion	Residual Reduction Tolerance
Pressure	V-Cycle	0.1	
X-Momentum	Flexible	0.1	0.7
Y-Momentum	Flexible	0.1	0.7
Z-Momentum	Flexible	0.1	0.7
Turbulent Kinetic Energy	Flexible	0.1	0.7
Turbulent Dissipation Rate	Flexible	0.1	0.7
Energy	F-Cycle	0.1	

Pressure-Velocity Coupling

Parameter	Value
Type	SIMPLE
Pseudo Time Method (Local Time-Step)	yes
Pseudo-Time Courant Number	10

Discretization Scheme

Variable	Scheme
Pressure	Second Order
Momentum	Second Order Upwind
Turbulent Kinetic Energy	First Order Upwind
Turbulent Dissipation Rate	First Order Upwind
Energy	Second Order Upwind

Solution Limits

Quantity	Limit
Minimum Absolute Pressure [Pa]	1
Maximum Absolute Pressure [Pa]	5e+10
Minimum Static Temperature [C]	-272.15
Maximum Static Temperature [C]	4726.85
Minimum Turb. Kinetic Energy [m^2/s^2]	1e-14
Minimum Turb. Dissipation Rate [m^2/s^3]	1e-20
Maximum Turb. Viscosity Ratio	100000